NAV-1 & NAV-2 | SUBTOPIC M-ST01A | SOURCE: Q1-Q5 BANK | PHYSICS TRUTH FIRST

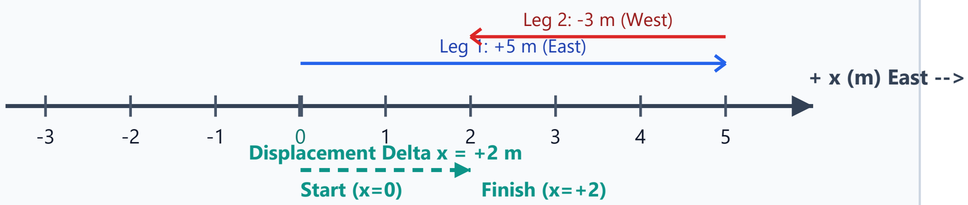
{{ LESSON_TITLE: Reference Frame, Coordinate Axis, Distance & Displacement }}

Core Pedagogy: SEE THE MOTION FIRST -> REALIZE THE VECTOR RELATION -> UNDERSTAND SIGN CONVENTIONS

{{ NAV_ROUTINE: Every Numerical Problem - First 20 Seconds }}

1. Draw a tiny coordinate line or functional sketch before touching equations.
2. Choose the positive direction (+ East / + Up) explicitly: sign errors stem from unchosen frames.
3. Translate hidden words into numerical physics data:
   'Starts from rest' => $u = 0$ | 'Comes to a stop' => $v = 0$ | 'Highest point' => $v_{top} = 0$.
4. Model Gate (NAV-2): Is acceleration constant? (Use UVATS) | Is velocity constant? (Use $s = vt$)
5. Avoid formula roulette: Select the single relation that directly bypasses unneeded unknowns.

{{ VISUAL_PRIMITIVE: 1D Coordinate Line & Return-Path Vectors }}



{{ PHYSICAL_INVARIANT: Path Length vs Coordinate Difference }}

* Distance (Scalar): Total cumulative path length travelled regardless of direction.
  Distance is always positive and strictly monotonically non-decreasing over time: $d >= 0$.
* Displacement (Vector): Net change in position comparing only endpoints:
  Delta $x = x_{final} - x_{initial}$. Displacement can be positive, negative, or zero.
* The Reversal Trap: Whenever an object reverses direction, Distance > |Displacement|.
* Magnitude Bound: |Displacement| <= Distance. Equality holds ONLY for unidirectional straight motion.

PHYSICS WORKED EXAMPLE

{{ EXAMPLE: Two-Leg Reversal Journey }}

Problem: A student starts at origin $x = 0$, walks 6 m East, then turns and walks 2 m West.
Calculate: (a) Total Distance, (b) Final Position, (c) Net Displacement.

Physics Reasoning Story (Words --> Picture --> Facts --> Target --> Relation):

1. Define Frame: Let East be positive (+x direction). Origin is the school gate $x = 0$ m.
2. Leg 1 Vector: Delta $x_1 = +6$ m (East). Position after Leg 1: $x_1 = 0 + 6 = +6$ m.
3. Leg 2 Vector: Moves West => Delta $x_2 = -2$ m. Position after Leg 2: $x_2 = +6 + (-2) = +4$ m.
4. Calculate Distance: d = |Delta x_1| + |Delta x_2| = 6 m + 2 m = 8 m. (Path length accumulation)
5. Calculate Displacement: Delta $x = x_{final} - x_{initial}$ = (+4 m) - (0 m) = +4 m (East).
6. **Verification Check: |Delta x| = 4 m < 8 m = Distance. Reversal detected and verified. [PASS]**

GUIDED PRACTICE * TRY WITH ME

{{ GUIDED_PRACTICE_Q1 }}: Start at x = -2 m. Walk 5 m East, then 7 m West.

* Step 1: Mark initial coordinate x_0 = ___ m.
* Step 2: Displacement vector Delta x_1 = ___ m, Delta x_2 = ___ m.
* Step 3: Final coordinate $x_{final} = x_0$ + Delta x_1 + Delta x_2 = ___ m.
* Step 4: Total Distance d = |___| + |___| = ___ m.
* Step 5: Net Displacement Delta $x = x_{final} - x_0$ = ___ m.

{{ GUIDED_PRACTICE_Q2 }}: Can an athlete complete one lap of a 400 m circular track and have a displacement of 400 m? Explain using the definition of displacement.

Your physics explanation: ____

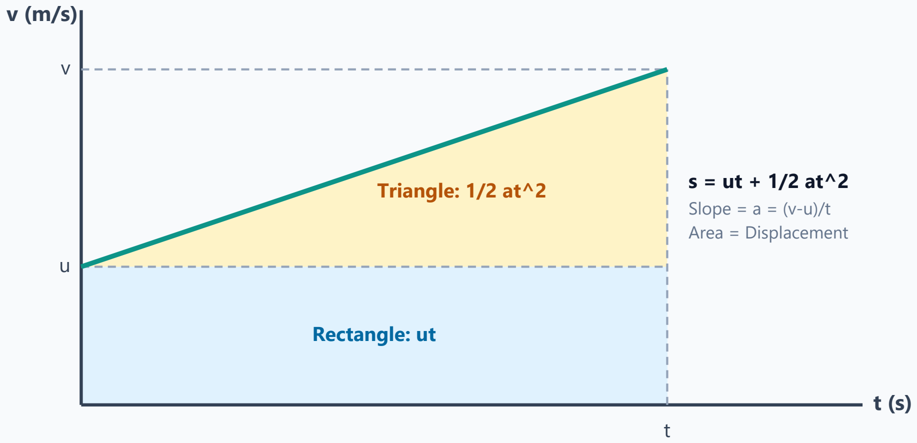
M-ST04A GRAPHS

SLOPE & AREA DECODER

SEE -> REALIZE -> UNDERSTAND

{{ LESSON_TITLE: Velocity-Time Area as Accumulated Movement & Kinematic Derivations }}

{{ VISUAL_PRIMITIVE: Constant Acceleration v-t Area Decomposition }}



GRAPH MISCONCEPTION CLINIC

{{ MISCONCEPTION: Graph Height vs Graph Slope }}

*** Misconception 1: 'Higher point on x-t graph means higher speed.'**

Correction: Height on an x-t graph indicates position (distance from origin).

Speed is the SLOPE (rate of change). A flat horizontal line at x = 100 m means v = 0.

*** Misconception 2: 'Area below zero velocity axis is still positive distance.'**

Correction: In v-t graphs, area below the v = 0 line represents NEGATIVE displacement

(travel in the negative coordinate direction).

To compute Total Distance: Add absolute areas |Area_above| + |Area_below|.

To compute Net Displacement: Take signed sum: Area_above - Area_below.

*** Model Validity Guardrail: The kinematic equations (s=ut+1/2at2, v2=u2+2as) are valid**

ONLY when acceleration is constant. For variable acceleration, area must be integrated.

{{ DERIVATION_WALKTHROUGH: Why s = ut + 1/2at2 Works Physically }}

1. Geometry of the Velocity-Time Profile:

Under constant acceleration 'a', velocity rises linearly from initial speed 'u' to 'v' in time 't'.

The region under the graph decomposes into a base rectangle and an upper triangle.

2. The Base Rectangle Area (Displacement if speed never increased):

Area_rect = base x height = t x u = ut. (The distance covered at steady speed u)

3. The Upper Triangle Area (Extra displacement gained from acceleration):

Area_tri = 1/2 x base x height = 1/2 x t x (v - u).

Since acceleration is defined as a = (v - u)/t => (v - u) = at.

Substitute: Area_tri = 1/2 x t x (at) = 1/2 at2.

4. Total Integrated Displacement: s = Area_rect + Area_tri = ut + 1/2 at2.

Physical Meaning: Initial movement + additional speed gain = total position shift.

CORE 1 -> CORE 2 TRANSFER PIPELINE

{{ CORE2_TRANSFER_ROADMAP: Practice Set Progression }}

You have now mastered the conceptual representations of 1D Motion.

In Core 2, you will encounter the audited PYQ & ExamSIDE transfer questions:

* Assimilation Set 1: Averages, Speed vs Velocity, Harmonic Mean
* Assimilation Set 2: Constant Acceleration Core (UVATS selection)
* Assimilation Set 3: Polynomial motion & braking vehicle pairs
* Assimilation Set 4: Free fall & vertical gravity stages
* Assimilation Set 5: Relative motion & common acceleration cancellation
* Assimilation Set 6: Velocity-Time & Position-Time graph interpretation

Transfer Rule: Always attempt before revealing H1-H3 hints. Check methods after finishing.